Motion and Forces

AI Tutor - Physics - Lesson 2

Describing Motion

Motion is a change in an object's position over time. We describe motion using speed (distance / time) and velocity (speed with direction).

Newton's Three Laws of Motion

- **First Law (Inertia)** - an object stays at rest or in uniform motion unless acted on by a force.
- **Second Law** - Force = mass × acceleration ($F = ma$).
- **Third Law** - for every action, there is an equal and opposite reaction.

Everyday Examples

A ball rolling on a flat floor eventually stops because friction (a force) acts against its motion. A rocket launches upward because the force of expelled gas pushes the rocket in the opposite direction (Newton's Third Law).